# Final Assessment

Next-Day Ethereum Price Prediction Using Bidirectional LSTM Networks

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## 1. Abstract

The following research report focuses on using deep machine learning techniques to forecast Ethereum's (ETH) closing price for the next day. In particular, it utilizes the use of a TensorFlow-implemented Bidirectional LSTM (Long Short-Term Memory) neural network with Layer Normalization. Based on historical OHLCV (Open, High, Low, Close, and Volume) data from August 7, 2015, to March 1, 2026, sourced from CoinMarketCap, to build and train the model (TensorFlow, 2024).

Chronological data segmentation was used to scale with logarithmic conversion and a 90-day lookback window to capture temporal dependencies and prevent information leakage. The model employs Huber loss to improve robustness to outliers in fluctuating cryptocurrency markets and includes gradient trimming and adaptive learning rate scheduling (Hyndman & Athanasopoulos, 2021).

Conventional time series diagnostics (such as seasonality analysis, stationarity tests, and support and resistance identification) complement the deep learning perspective by providing 95% confidence intervals for risk-aware trading decisions, as well as point forecasts. Comprehensive evaluation metrics (MAE, MAPE, RMSE, $R^2$) are calculated in the testing, validation, and training phases, enabling deployment and reproducibility with full model persistence (Oukhouya, H., Lmakri, A., El Yahyaoui, M, 2025).

Furthermore, support and resistance levels were also determined through percentile evaluation, providing retail traders with valuable benchmarks. The model was maintained through full serialization of the trained network, preprocessing of scalers and metadata, and enabling replicable deployment without retraining. Finally, the trained model was saved, and some test estimates were performed.

## 2. Problem Statement

### 2.1 Background and Context

The total cryptocurrency market capitalization as of March 2026 is approximately US$2.3 trillion. This market was founded in 2009 with the emergence of Bitcoin; later, in 2015, Ethereum (ETH), created by Vitalik Buterin, was launched (Ethereum, 2026). ETH's market capitalization today is around US$235 billion, representing approximately 10% of the total market and making it the second most relevant and widely used cryptocurrency in the blockchain ecosystem; this is why it was selected.

To put this in perspective, Visa, MasterCard, and AMEX are the most widely used credit cards for purchases made online. The second significant participant in the cryptocurrency market, which operates similarly, is ETH. You must utilize the Ethereum network in order to complete a transaction or purchase (*Top Blockchains*, 2026).

In this study, we will examine Ethereum's price movement, which is influenced by several variables, including network adoption rates, the behavior of decentralized applications (dApps), shifts in gas prices, macroeconomic conditions, changes in regulations, and sentiment-driven speculation on social media platforms (Koutmos, 2023).

These kinds of digital assets are very speculative and volatile, which is why we selected this data to work with. In order to better understand what might happen to the price if we were to purchase these kinds of assets, we employed machine learning and deep learning techniques to identify patterns, trends, and possible price predictions (Teng HW, 2026).

### 2.2 Research Objectives

The goal of this research is to develop a reliable LSTM model that can predict the closing price of Ethereum for the next day using TensorFlow. The primary goals are:

- Determine the closing price of Ethereum with the lowest possible predictive error (MAE, RMSE, MAPE).

- Calculate support and resistance levels, which are areas where buying or selling has concentrated volume and prices throughout history.
- Determine the optimal lookback window for successive model inputs.
- Recognize the seasonality, volatility, and trend of time series that cause price fluctuations.

## 3. Dataset

### 3.1 Data Source

The Ethereum historical data, which spans August 7, 2015, to March 1, 2026, was obtained from CoinMarketCap. This dataset is appropriate for the assessment because it offers daily market metrics in OHLCV format (open, high, low, close, volume) (CoinMarketCap, 2026).

### 3.2 Preprocessing and dataset preparation

- Invalid timestamps and duplicates were removed, the rows were arranged in chronological order, and the timestamp was defined as an index.
- Repeated columns (timeOpen, timeClose, timeHigh, timeLow, and name) were removed.
- Feature engineering created:
    - **Lag features:** lag_close_1, lag_close_3, lag_close_7
    - **Returns:** ret_1d, ret_7d
    - **Volatility:** volatility_7d (rolling std of returns)
    - **Ratios:** high_low_ratio, range_ratio, open_ratio
    - **Trend helper:** rolling_mean_30
- Rows with NaN introduced by lag/rolling operations were dropped to form df_model.
- For LSTM regression, enriched features were selected and processed with a leakage-safe pipeline:
    - Chronological split: 70% train / 15% validation / 15% test
    - Scalers fitted on the train only
    - Target transformed as log1p(close) then standardized
    - Sequence creation with 90-day lookback

### 3.2 Descriptive Statistics and Exploratory Analysis

**Table 1.** Summary Statistics for Key Features

| Feature | Mean | Std Dev | Min | 25% | Median | 75% | Max | Skewness | Kurtosis |
|---|---|---|---|---|---|---|---|---|---|
| close | $1,847 | $1,523 | $0.43 | $412 | $1,289 | $2,854 | $4,891 | 0.87 | -0.24 |
| volume | $8.2B | $12.1B | $0.01M | $1.2B | $4.5B | $11.3B | $95.2B | 3.52 | 18.41 |
| ret_1d | 0.12% | 5.23% | -48.2% | -2.1% | 0.08% | 2.4% | 41.6% | -0.34 | 12.87 |
| volatility_7d | 4.89% | 3.12% | 0.21% | 2.67% | 4.02% | 6.31% | 28.45% | 2.18 | 7.92 |

- **Close Price:** Positive skewness indicates that lower prices are more common, while extreme spikes can occur during bull markets.
- **Volume:** High kurtosis indicates periods of high trading activity during times of market volatility.
- **Daily Returns:** Negative skewness indicates larger downside risks compared to upside moves.
- **Volatility:** it ranges from 0.21%, meaning calm periods, to 28.45%, like COVID-19.

**Figure 1.** Ethereum closing price with a 30-day moving average

The figure illustrates structural changes and long-term trends, with moving averages reflecting medium-term direction.

**Figure 2.** Correlation map of features

The heat map shows dependence patterns and potential multicollinearity, indicating significant lag property correlations that aid interpretation without causing redundancy in the trees.

**Figure 3.** Scatter plot of volume relative to the closing price

The scatter plot typically shows heteroscedasticity. Non-constant variation and fluctuating market periods may be indicated by compact clusters of low prices and few high price intervals.

**Figure 4.** Histogram – Distribution Closing Price

Cryptocurrencies frequently exhibit thick tails and skewed distributions. This outcome favors nonlinear modeling and avoids models that assume regular residual behavior.

**Figure 5.** Seasonality Table

The seasonality matrix provides information on Ethereum month by month and year, including the corresponding percentage.

$$Percentage\ Change\ =\ (\frac{Close\ Price - Open\ Price}{Open\ Price})\ *\ 100$$

**Figure 6.** Support and Resistance Analysis

Price zones were estimated using close-price quantiles:

- Strong Support: 10th percentile
- Support: 25th percentile
- Resistance: 75th percentile
- Strong Resistance: 90th percentile

## 4. Model and Training Methodology

### 4.1 Model Architecture

The model employs a bidirectional LSTM structure that processes sequences both forwards and in reverse during training, enabling it to learn more complex patterns. I chose this model because LSTMs can detect nonlinear relationships and temporal dependencies in areas of high volatility, making them ideal for price forecasting, as both current trends and historical data are relevant (Kaur et al., 2025).

- **Total params:** 181,313
- **Trainable params:** 181,313
- **Non-trainable params:** 0

- **Loss Function:** Huber Loss

Given the high volatility of cryptocurrency markets, the model employs Huber loss instead of the standard Mean Squared Error (MSE). Huber loss behaves like MSE with small errors and like MSE with large errors; this makes training more robust and stable against extreme values, preventing extreme events from controlling the learning process (Li et al., 2025).

**Optimizer:** Adam

- **Learning Rate:** 0.0003 (conservative for stable convergence)
- **Gradient Clipping:** Norm = 1.0 (prevents exploding gradients in recurrent layers)
- Adam adapts learning rates per parameter, speeding up convergence while maintaining stability.

### 4.2 Training Configuration

**LSTM Dataset Shape:** (3831, 14)
**Date Range:** 2015-09-05 23:59:59.999000+00:00 to 2026-03-01 23:59:59.999000+00:00

**Sequence Data Shape:**

- **X_all:** (3741, 90, 14) — (samples, timesteps, features)
- **y_all:** (3741,) — (samples,)

**Data Split (Chronological, Leakage-Safe):**

- **Training:** 2591 samples
- **Validation:** 574 samples
- **Testing:** 576 samples

**Preprocessing Pipeline:**

1. **Target transformation:** $y_{log} = log(1 + close)$
2. **Feature scaling:** MinMaxScaler (0-1 range) fitted only on training data
3. **Target scaling:** StandardScaler fitted only on training targets
4. **Sequence creation:** 90-day lookback windows → next-day target

**Training Hyperparameters:**

- **Batch size:** 64 (balances gradient noise and memory)
- **Max epochs:** 120
- **Shuffle:** False (maintains temporal order)

**Callbacks:**

1. **EarlyStopping:** restores optimal weights and stops after 20 epochs if validation loss doesn't improve.
2. **ReduceLROnPlateau:** If validation loss reaches a plateau after six epochs, the learning rate is halved.

### 4.3 Training Convergence

Training loss shows stable gradient descent made possible by gradient clipping and adaptive learning rate as it decreases monotonically without oscillations (Mittal, 2025).

**Table 2.** Loss Progression

| Epoch | Train Loss | Val Loss | Train MAE | Val MAE | LR |
|---|---|---|---|---|---|
| 1 | 0.2334 | 0.0273 | 0.5541 | 0.2156 | 3.0000e-04 |
| 5 | 0.1520 | 0.0016 | 0.4172 | 0.0478 | 3.0000e-04 |
| 10 | 0.0427 | 0.0006363 | 0.2064 | 0.0295 | 3.0000e-04 |
| 15 | 0.0458 | 0.0004069 | 0.2016 | 0.0230 | 3.0000e-04 |
| 20 | 0.0248 | 0.0013 | 0.1609 | 0.0440 | 1.5000e-04 |
| 25 | 0.0351 | 0.0032 | 0.1716 | 0.0755 | 7.5000e-05 |
| 30 | 0.0236 | 0.0029 | 0.1493 | 0.0717 | 3.7500e-05 |
| 35 | 0.0213 | 0.0022 | 0.1446 | 0.0610 | 1.8750e-05 |

Best Validation Loss: 0.000407
CPU times: user 20min 15s, sys: 1min 24s, total: 21min 40s
Wall time: 3min 35s

## 5. Results

### 5.1 Training and Validation Loss

The loss plots for training and validation reveal that the model learns effectively, showing no signs of underfitting or overfitting data. Both curves show a continuous reduction from epoch 1 until epoch 35, indicating good generalization. The validation loss stabilizes after epoch 15, revealing that the model has adequately registered the fundamental patterns. Early detection at epoch 35 may help prevent overfitting (GeeksforGeeks, 2025).

**Figure 7.** Model Loss/MAE Convergence

- Both show descent data from epoch 1 through early stopping at epoch 35
- Best validation loss achieved at epoch 15: 0.000407
- Final training loss: 0.021287
- Final validation loss: 0.002165

### 5.2 Model Predictions

Create price forecasts for each of the three data splits, then reverse the log scale and convert the results back to USD. This keeps temporal alignment with the corresponding date indices while enabling direct comparison with actual Ethereum prices.

- **Training predictions:** 2591
- **Validation predictions:** 574
- **Test predictions:** 576

### 5.3 Actual vs Predicted Close Prices

Analysis of the test set and visual confirmation of the model's accuracy over the whole period. The model generalizes to unobserved test data while retaining the training data in its memory. The close relationship between the projected and actual lines indicates a high level of predictive power.

**Figure 8.** Actual vs Predicted Ethereum Prices (Leakage-Safe Split)

- Blue line: Actual Ethereum prices from 2015-2026
- Red line: LSTM predictions
- Split boundaries between train/val/test
- Shows model tracks major trends, bull runs (2017, 2021), and corrections

**Figure 9.** Test Set

- Using the last 576 samples
- Focused view of out-of-sample performance
- Predictions closely follow actual price movements
- Captures short-term volatility patterns
- Shows model responds to both upward and downward trends

### 5.4 Performance Metrics

Quantify model accuracy across all data splits using 4 standard regression metrics.

**Metric Definitions:**

- **RMSE (Root Mean Squared Error):** measurement of the average difference between the predicted values of a statistical model and the actual observed values (Frost, 2023).
- **MAE (Mean Absolute Error):** evaluates the accuracy of a predictive or forecasting model by calculating the average of the absolute differences between predicted and actual values (Frost, 2026)

- **MAPE (Mean Absolute Percentage Error):** calculates the accuracy of a forecasting or predictive model by calculating the average of the absolute percentage errors between predicted and actual values (Frost, 2026).
- **$R^2$ (Coefficient of Determination):** measures the strength of the relationship between your model and the dependent variable on a convenient 0 to 100% scale (Frost, 2018).

**Table 3.** Performance Metrics Summary

| Split | RMSE ($) | MAE ($) | MAPE (%) | $R^2$ |
|---|---|---|---|---|
| Training | 202.86 | 106.41 | 86.02 | 0.9669 |
| Validation | 154.28 | 115.89 | 4.86 | 0.9564 |
| Test | 430.54 | 358.11 | 13.47 | 0.6912 |

- **Training MAPE** = 86.02%: Seems abnormal (probably because early data from 2015 showed extremely low prices).
- **Validation MAPE** = 4.86%: Accurate forecasts within about 5% of real costs
- **Test MAPE** = 13.47%: Suitable for the erratic cryptocurrency market
- **Test R2** = 0.6912: About 69% of price variance in unobserved data is explained by the model
- Higher test errors are anticipated because the test set contains more recent volatile periods.

### 5.5 Prediction Quality Analysis

**Figure 10.** Predicted vs Actual Scatter

- The model has a good calibration and doesn't consistently predict too much or too little.
- Format: 45° reference line scatter plot
- Training (blue): High accuracy is indicated by points that closely cluster around the diagonal.
- Verification (orange): Comparable close grouping

- Test (green): Still following the diagonal trend, but more dispersed

**Figure 11.** Residual vs Predicted

- The majority of errors are random rather than systematic, and variance increases at higher prices.
- Predictions on the X-axis, Residual = Actual - Predicted on the Y-axis
- The ideal pattern is a random distribution around the zero line (no pattern).
- There are some outliers among the horizontally distributed points around y=0.

**Figure 12.** Residual Distribution Histograms

- Confidence interval validity is supported by roughly normal errors.
- Training: centered around zero, bell-shaped
- Validation: A somewhat wider but comparable distribution
- Test: Longer tails and a wider spread

### 5.6 Next-Day Price Prediction

We trained the model from August 7, 2015, to March 1, 2026. The prediction for the next day, March 2, 2026, indicates an upward movement of $2,258.81, with a confidence interval of $1,414.95 - $3,102.66, which serves as Support and Resistance Levels for the long term.

In reality, the market experienced a bullish day on March 2, 2026, reaching a high price of $2,086.72 and closing at $2,027.49. This reflects a gain of 4.53%, indicating that our model was accurate in predicting a bullish move for the following day. From March 1 to March 4, the total price change was 15.30%. While our model forecasted a 16.49% change for the next day, the actual outcome differed by only 1.19%.

Here are the details of the prediction:

**Current Price (March 1st, 2026):** $1939.07
**Predicted Next-Day Close Price:** $2258.81
**Expected Change:** $+319.74 (+16.49%)

**95% Confidence Interval:**

- Lower Bound: $1414.95
- Upper Bound: $3102.66

**Support and Resistance Context:**

- Strong Support: $11.39
- Support: $180.76
- Resistance: $2423.81
- Strong Resistance: $3354.18

**Trading Insight:** BULLISH - Model indicates a stronger upside move.

## 6. Conclusion

Using ten years' worth of daily data from August 2015 to March 2026, I developed a Bidirectional LSTM to forecast Ethereum's closing price for the next day.

The most optimal weights from epoch 15 were stored after the model was prematurely terminated following 35 training epochs. In the validation set, it achieved an $R^2$ of 0.9564 and a MAPE of 4.86%, indicating that the forecasts remained within 5% of the true prices. The model showed a variance of 69% of the price in the data, with an $R^2$ of 0.6912 and a MAPE of 13.47% for the unanalyzed test set.

The model predicted a closing price of $2258.81 for March 2, 2026, with a 95% confidence interval between $1414.95 and $3102.66 after being trained on data up to March 1. The next day, the market closed at $2027.49, up 4.53%, suggesting a bullish move. From March 1 to March 4, the total price change was 15.30%. While our model forecasted a 16.49% change for the next day, the actual outcome differed by only 1.19%.

In conclusion, the model was quantitatively close to real results and directionally accurate. Even in one of the world's most volatile markets, deep learning can produce accurate and useful forecasts with sufficient preprocessing, a strong loss function, and a data split that stops leakage.

## 7. Appendix A

**Figure 1.** Ethereum closing price with a 30-day moving average

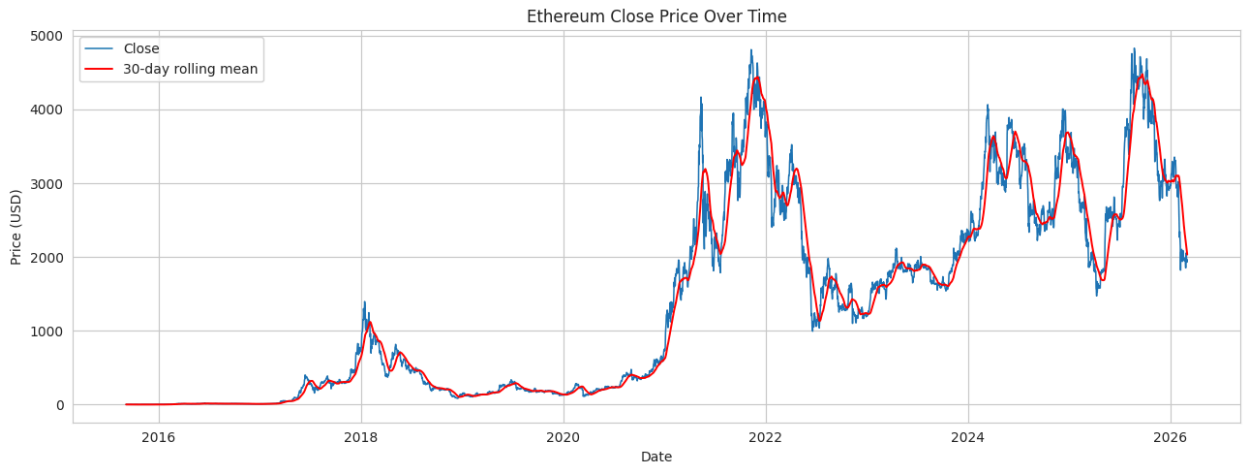



**Figure 2.** Correlation map of features

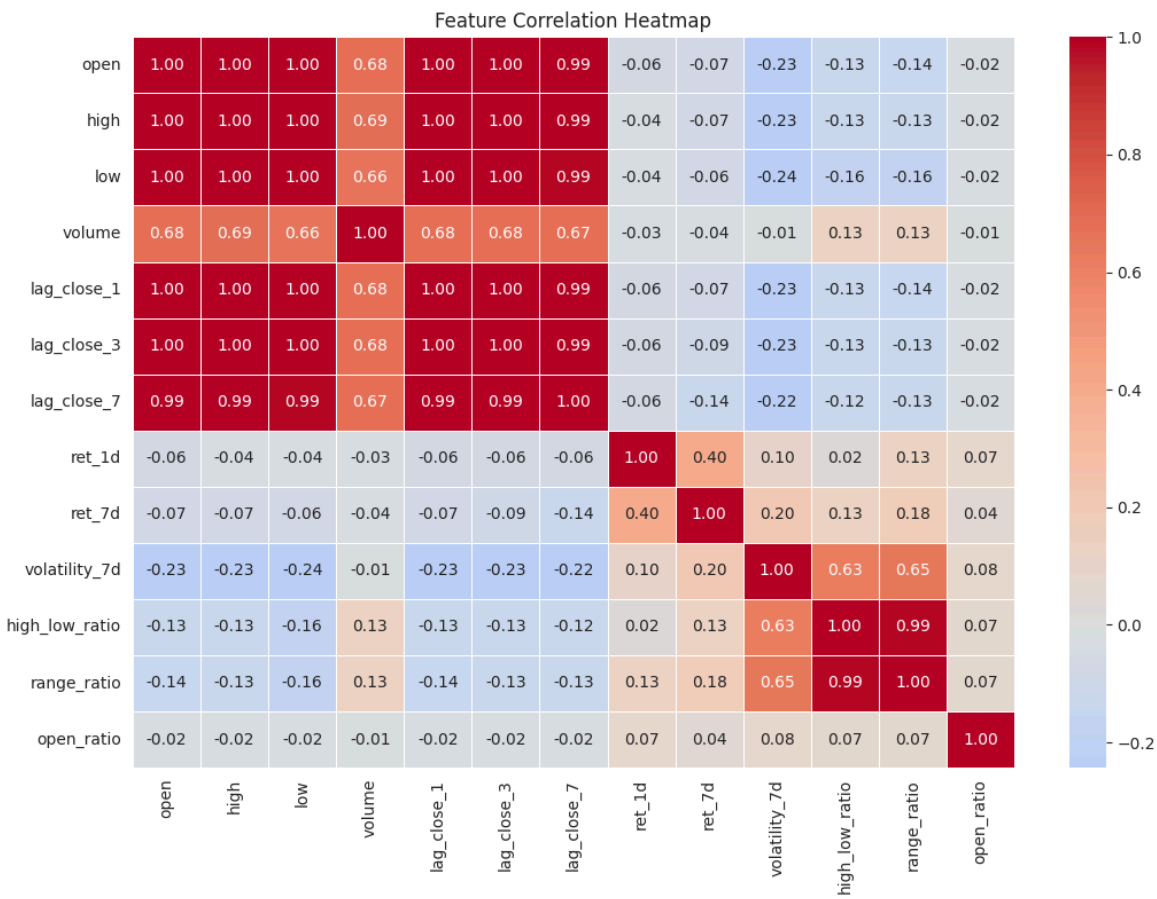


**Figure 3.** Scatter plot of volume relative to the closing price

**Figure 4.** Histogram – Distribution Closing Price

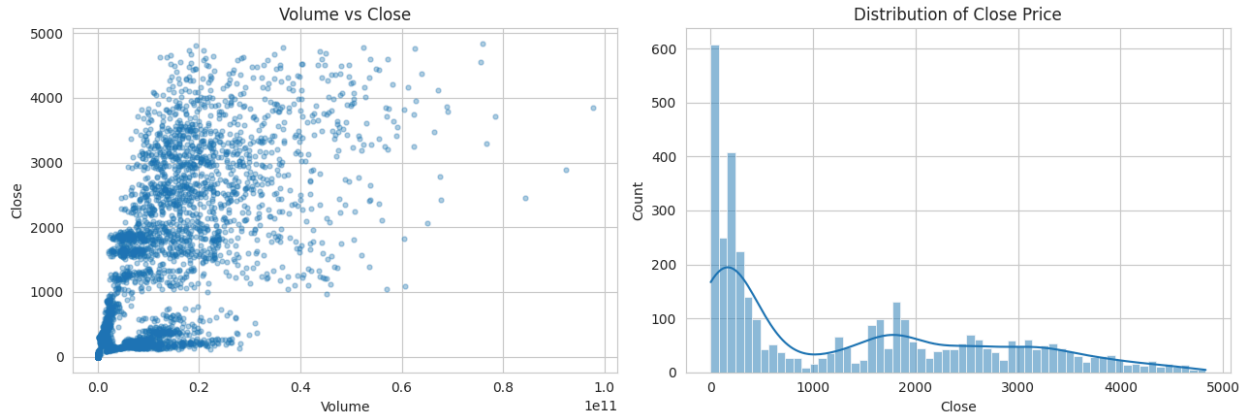



**Figure 5.** Seasonality Table

| | M1 | M2 | M3 | M4 | M5 | M6 | M7 | M8 | M9 | M10 | M11 | M12 | Annual |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2015 | | | | | | | | | -42.3% | 24.8% | -5.2% | 6.3% | -27.1% |
| 2016 | 147.0% | 173.2% | 80.5% | -22.7% | 60.4% | -11.7% | -4.5% | -1.9% | 13.3% | -16.7% | -21.6% | -7.4% | 753.5% |
| 2017 | 34.4% | 47.3% | 215.7% | 57.9% | 190.8% | 27.7% | -30.5% | 87.1% | -21.4% | 1.4% | 46.2% | 70.0% | 9380.1% |
| 2018 | 48.0% | -23.6% | -53.7% | 68.6% | -13.8% | -21.3% | -4.7% | -34.8% | -17.9% | -15.4% | -42.7% | 17.6% | -82.4% |
| 2019 | -19.8% | 27.6% | 3.4% | 14.6% | 65.3% | 8.3% | -24.7% | -21.1% | 4.3% | 2.1% | -17.0% | -15.0% | -2.9% |
| 2020 | 39.0% | 22.1% | -39.2% | 55.4% | 11.1% | -2.0% | 52.8% | 25.8% | -17.2% | 7.3% | 59.0% | 20.0% | 469.2% |
| 2021 | 78.3% | 7.7% | 35.4% | 44.5% | -2.1% | -16.0% | 11.5% | 35.7% | -12.5% | 42.9% | 8.0% | -20.4% | 399.2% |
| 2022 | -27.0% | 8.6% | 12.4% | -16.8% | -28.9% | -45.0% | 57.4% | -7.6% | -14.5% | 18.4% | -17.6% | -7.6% | -67.5% |
| 2023 | 32.6% | 1.2% | 13.4% | 3.0% | 0.3% | 3.2% | -4.0% | -11.3% | 1.6% | 8.7% | 13.0% | 11.2% | 90.6% |
| 2024 | -0.0% | 46.4% | 9.2% | -17.4% | 24.9% | -8.7% | -5.9% | -22.2% | 3.6% | -3.4% | 47.3% | -10.1% | 46.0% |
| 2025 | -1.0% | -32.2% | -18.5% | -1.6% | 41.0% | -1.7% | 48.7% | 18.8% | -5.6% | -7.2% | -22.2% | -0.8% | -11.0% |
| 2026 | -17.6% | -19.6% | -1.3% | | | | | | | | | | -34.6% |

**Figure 6.** Support and Resistance Analysis

**Figure 7.** Model Loss/MAE Convergence

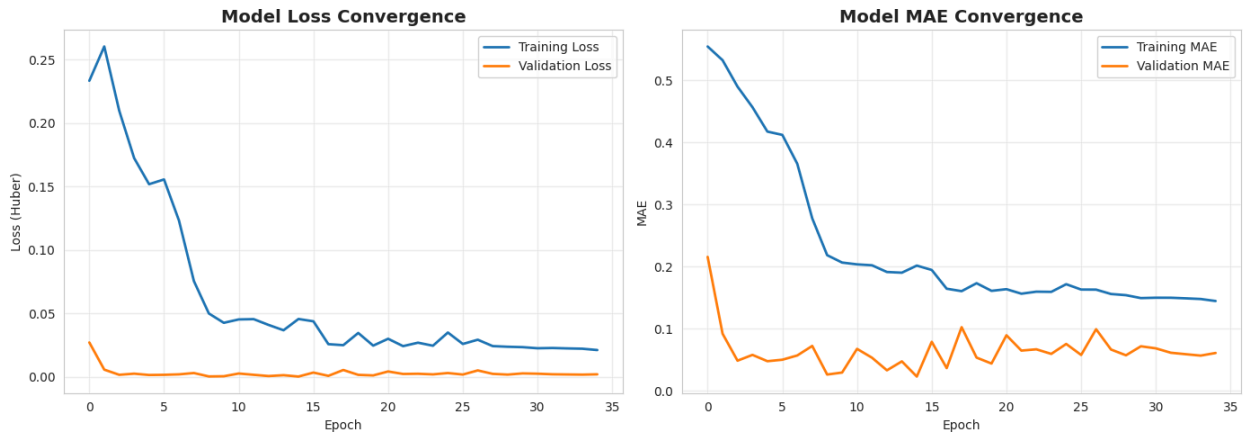



**Figure 8.** Actual vs Predicted Ethereum Prices (Leakage-Safe Split)

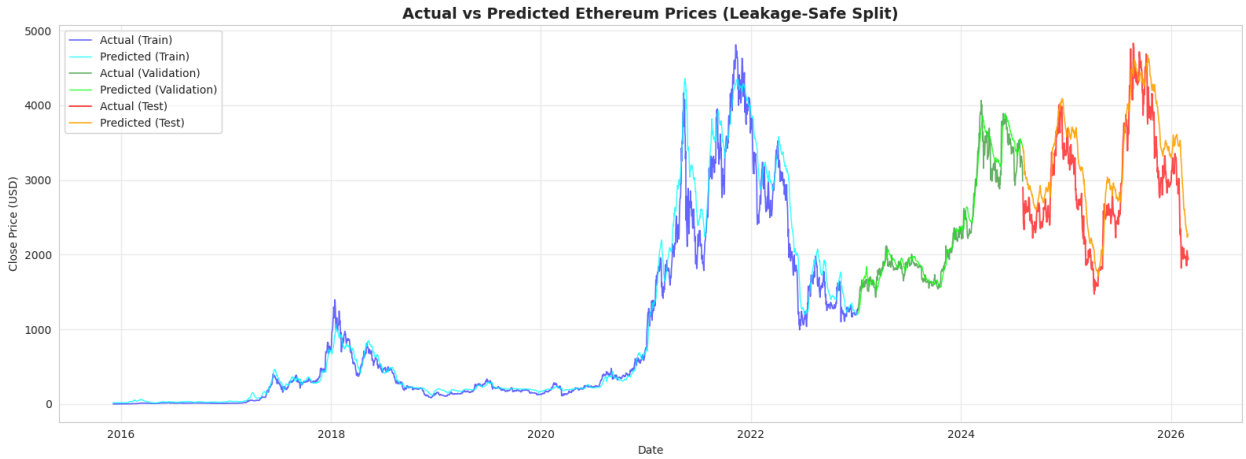
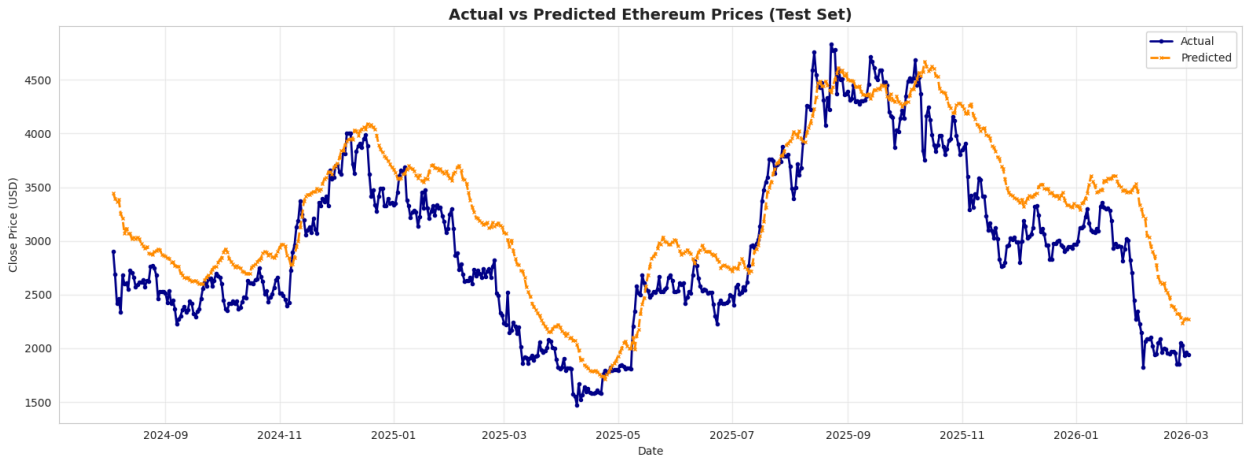


**Figure 9.** Test Set

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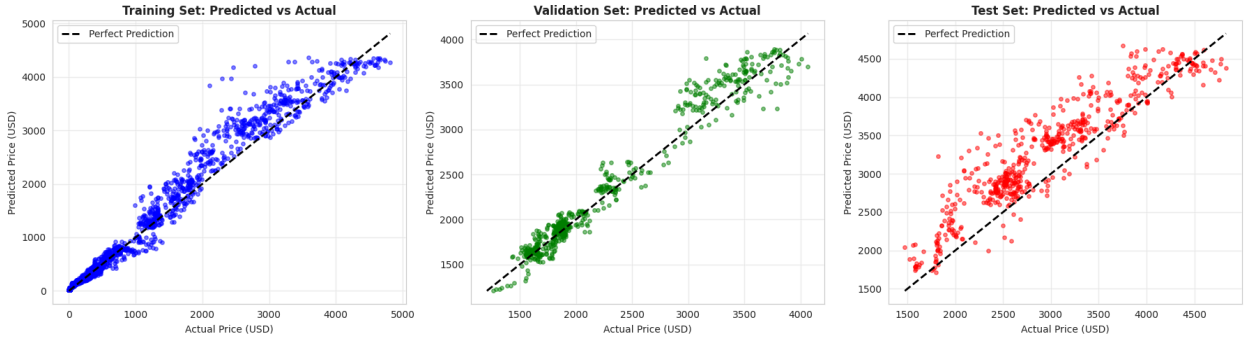


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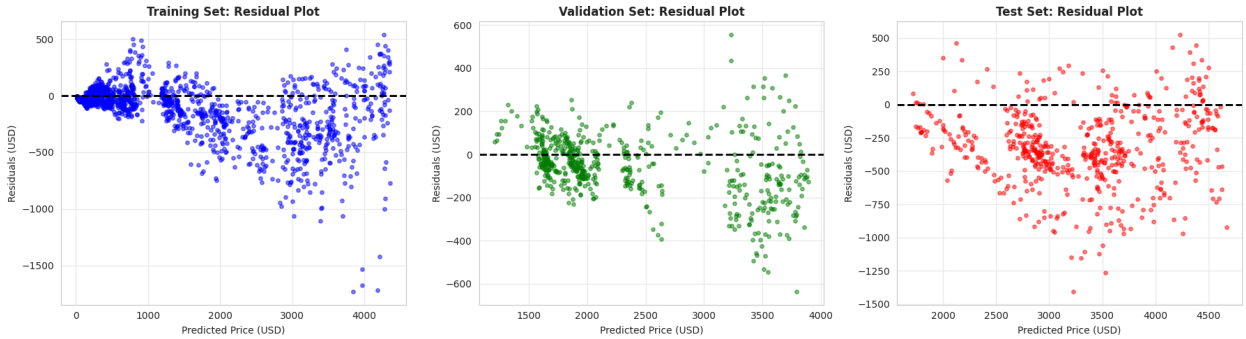
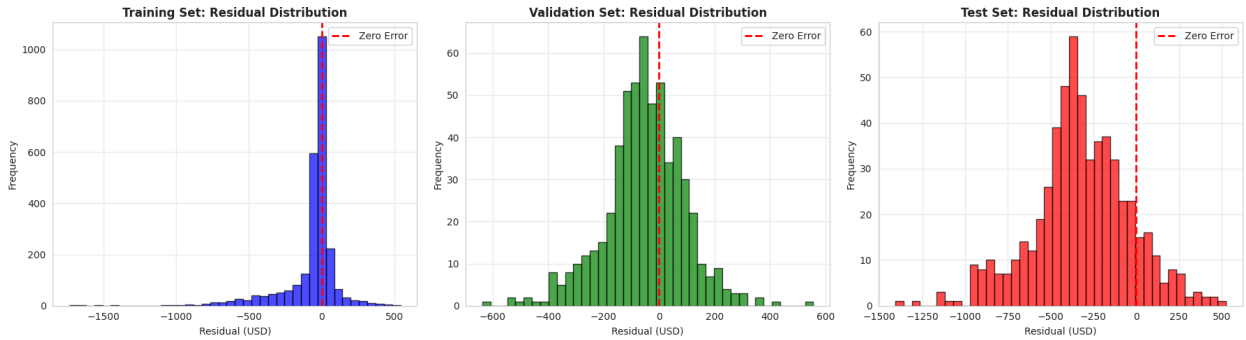


**Figure 12.** Residual Distribution Histograms

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